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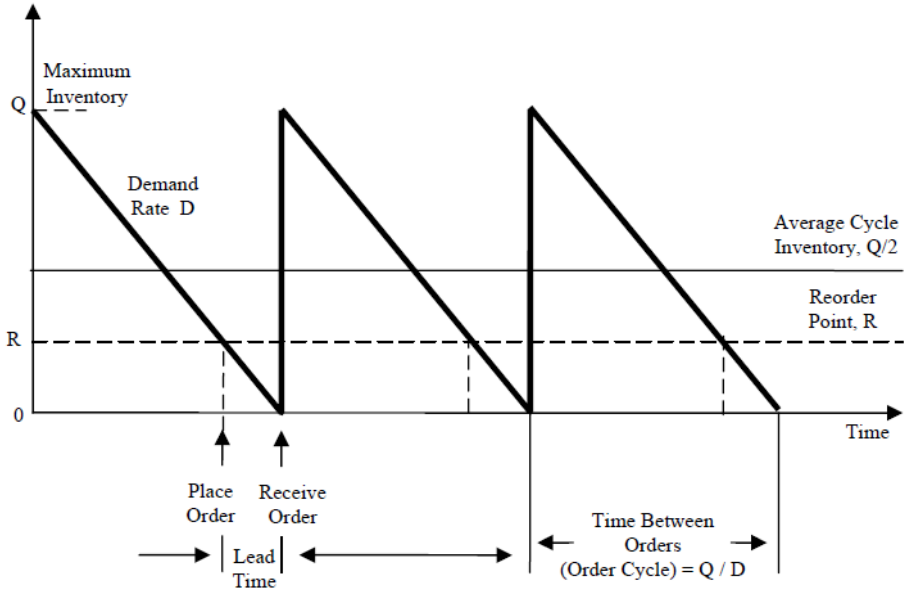
MZ 505
Inventory Systems

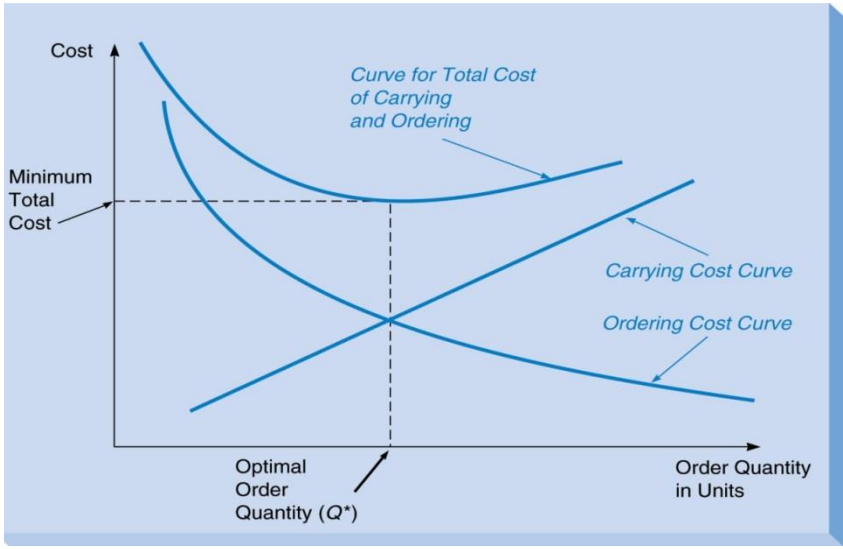
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Session 03:
EOQ Model Extensions

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EOQ model review





$$Q^* = \sqrt{\frac{2DS}{H}}$$

$$TVC(Q^*) = \sqrt{2DSH}$$

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What if we relax some of the EOQ model assumptions?

- ✓ Demand for the product is known and constant throughout the planning horizon
- ✓ Lead time (time from ordering to receipt) is known and constant
- ✓ Receipt of inventory is instantaneous
- ✓ Price per unit of product is constant
- ✓ Inventory holding cost is based on average inventory
- ✓ Ordering or setup costs are constant
- ✓ All demands for the product will be satisfied (no back orders are allowed)

EOQ –
FINITE PRODUCTION RATE

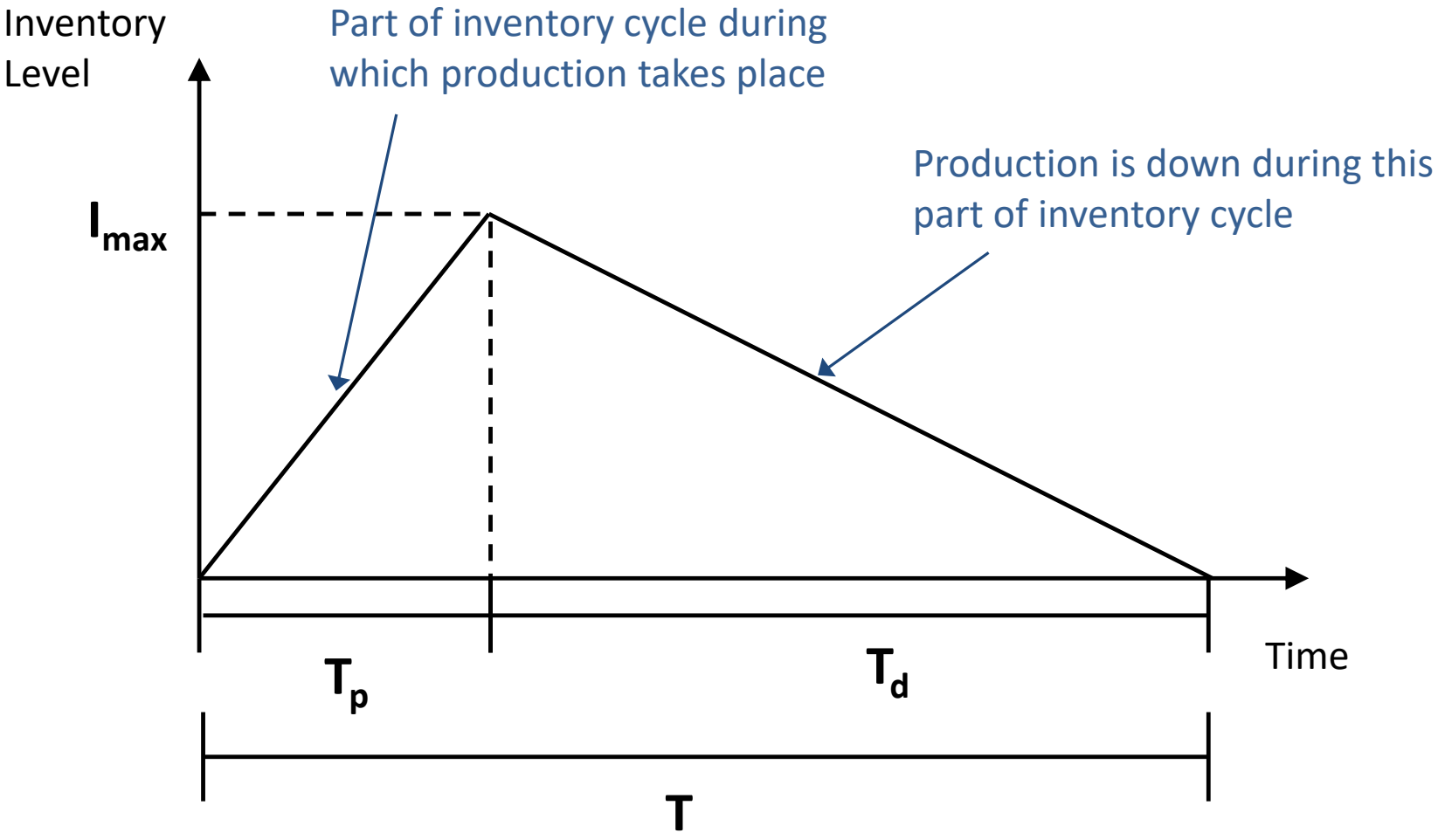
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Economic Production Quantity (EPQ)

- The EOQ model assumes that inventory arrives instantaneously in batches. In many cases though inventory arrives gradually.
  - In production environments, you may need to consider a finite production rate.
- The **economic production quantity (EPQ)** model assumes that inventory is being produced at a rate of P product units per time unit (P>D)
  - There is a setup cost (S) each time production begins
- Additional notation
  - T_p = production time per cycle
  - T_d = down time (non-production) per cycle
  - I_{max} = maximum inventory

EPQ dynamics



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Determining lot size or EPQ

Parameters

Q = Production quantity
S = Setup cost
D = Demand rate
P = Production rate
H = Holding cost per product unit per time unit

Computing maximum inventory (I_{max})

$$T_p = \frac{Q}{P}$$

$$I_{max} = T_p(P - D) = Q\left(1 - \frac{D}{P}\right)$$

$$T_d = \frac{I_{max}}{D} = \frac{Q}{D}\left(1 - \frac{D}{P}\right)$$

$$T = T_p + T_d = \frac{Q}{D}$$

EPQ model: Finding Q*

Total setup cost = (D/Q) x S

Total holding cost = (I_{max} /2) x H

} Total Variable Costs

Total production cost = C x D (where C here is the unit production cost)

Total variable cost: $$TVC(Q) = \left(\frac{D}{Q}\right)S + \frac{HQ}{2}\left(1 - \frac{D}{P}\right)$$

Note:

- (I_{max}/2) is the average inventory level
- As in EOQ, production cost does not depend on Q

Optimal EPQ and TVC (for P>D):

$$EPQ = Q^* = \sqrt{\frac{2DS}{H\left(1 - \frac{D}{P}\right)}}$$

$$TVC(Q^*) = \sqrt{2DSH\left(1 - \frac{D}{P}\right)}$$

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EOQ –
QUANTITY DISCOUNTS

What if we relax some of the EOQ model assumptions?

- ✓ Demand for the product is known and constant throughout the planning horizon
- ✓ Lead time (time from ordering to receipt) is known and constant
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- ✓ All demands for the product will be satisfied (no back orders are allowed)

Quantity discount models

- A **quantity discount** is a reduced unit price based on purchasing a large quantity (i.e., material cost displays economies of scale)
- Example discount schedule:

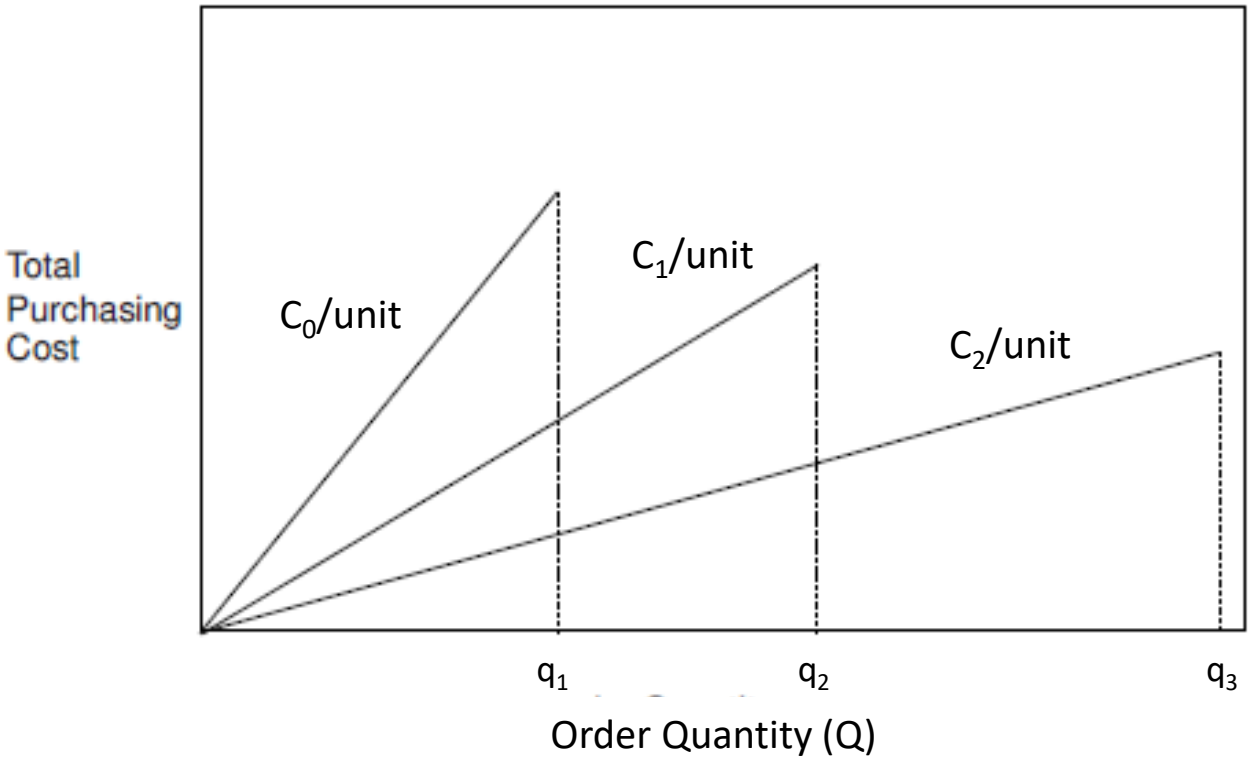
Pricing Schedule	Quantity	Discount	Unit Cost
1	0 to 1000	0%	$5.00
2	1,001 to 2,000	3%	$4.85
3	2,001 and over	5%	$4.75

- Lot size-based discount
  - All unit quantity discount
  - Marginal unit quantity discount
- Volume-based discount

All unit quantity discounts

- The pricing schedule contains specified break points q_0, q_1,…,q_m, where usually $q_0=0$ and $q_m=\infty$
- If the order Q is between q_i and q_{i+1}, each unit (in the entire order) is obtained at a cost C_i
- In general, the unit cost decreases as the quantity ordered increase; i.e., $C_0 \geq C_1 \geq \cdots \geq C_{m-1}$
- The objective for the company is to decide on a lot size that will minimize the sum of material, order, and holding costs
  - **TVC(Q) must now also include the purchasing cost**

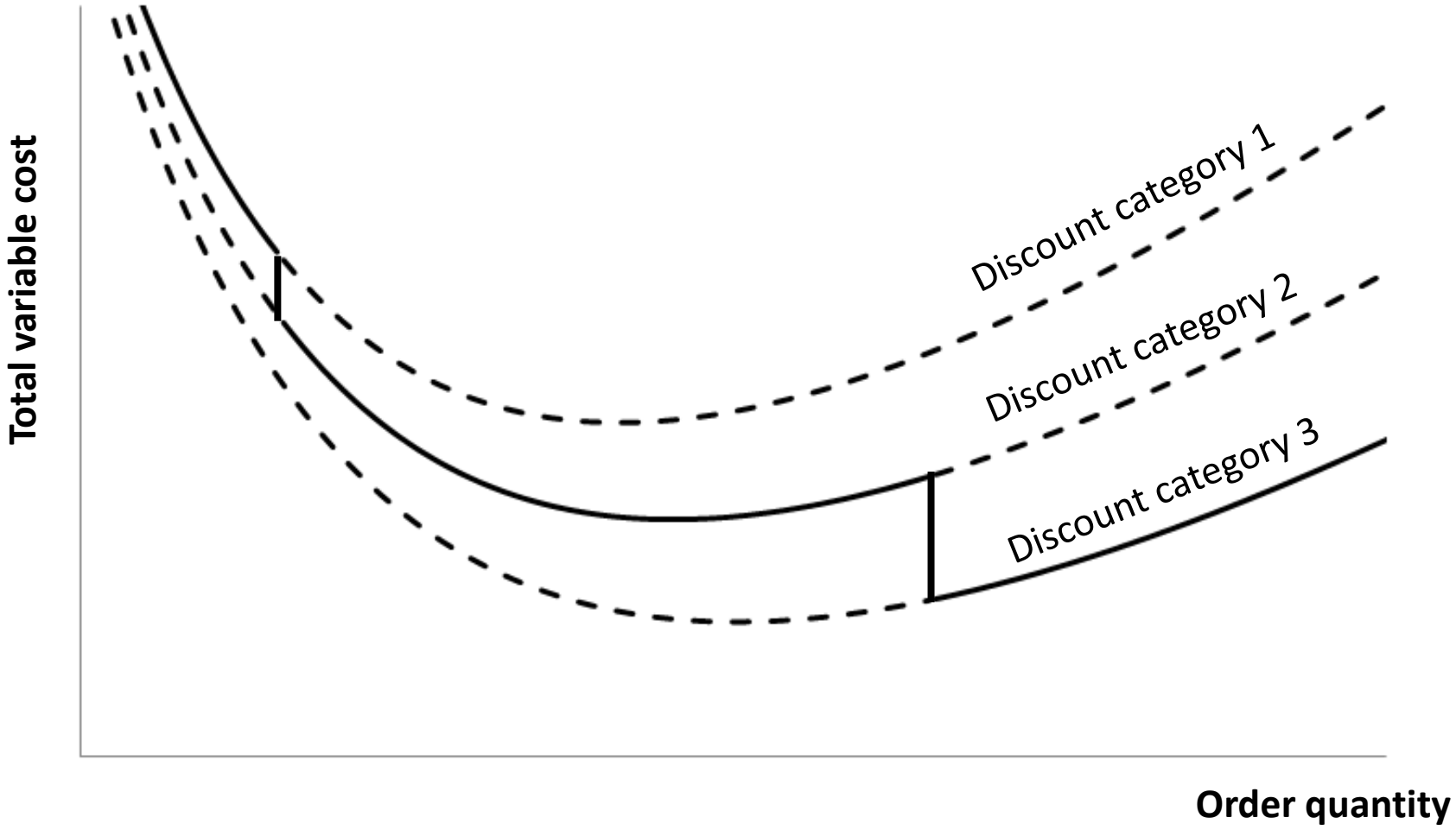
All unit quantity discounts: TVC function



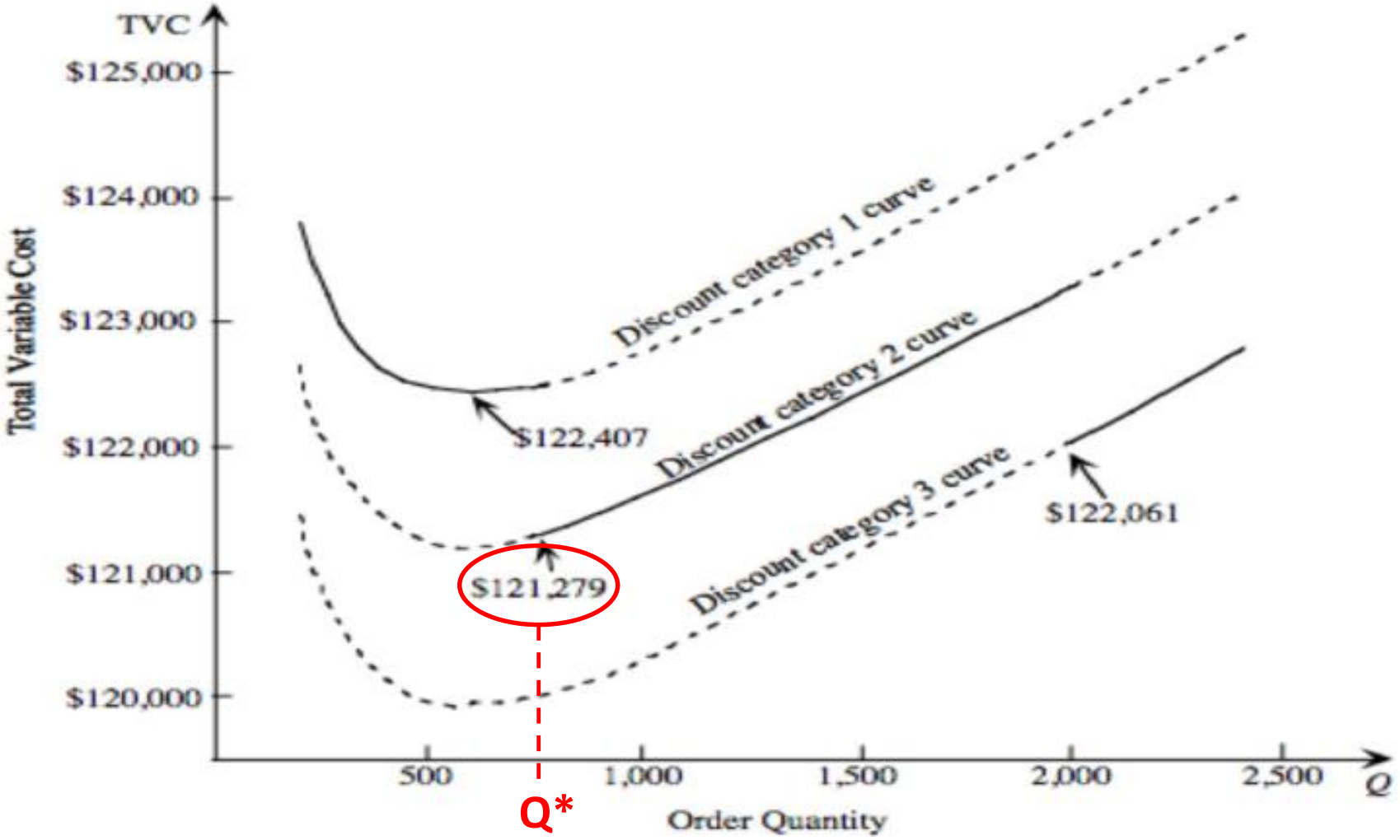
$$TVC_i(Q) = \frac{D}{Q}S + \frac{Q}{2}hC_i + DC_i \text{ , for } q_i \leq Q < q_{i+1}$$

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All unit discount: TVC vs. Q



All unit discount: Finding the EOQ (example)



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All unit quantity discounts: Solution algorithm

Step 1: Calculate the EOQ for the lowest price. If it is feasible (i.e., this order quantity is within the range for that price), then stop. This is the optimal lot size. Calculate total variable cost (TVC) for this lot size.

Step 2: If the EOQ is not feasible, calculate the TVC for this price and the smallest quantity for that price.

Step 3: Calculate the EOQ for the next lowest price. If it is feasible, stop and calculate the TVC for that quantity and price.

Step 4: Compare the TVC for Steps 2 and 3. Choose the quantity corresponding to the lowest TVC.

Step 5: If the EOQ in Step 3 is not feasible, repeat Steps 2, 3, and 4 until a feasible EOQ is found.

All unit quantity discounts: Alternative approach

- Starting from the lowest price, evaluate the EOQ for **each** C_i :

$$Q_i = \sqrt{\frac{2DS}{hC_i}}$$

  - Case 1: If $q_i \leq Q_i < q_{i+1}$, then Q_i is feasible and

$$TVC_i = \frac{D}{Q_i}S + \frac{Q_i}{2}hC_i + DC_i$$

  - Case 2: If $Q_i < q_i$, then choose q_i and

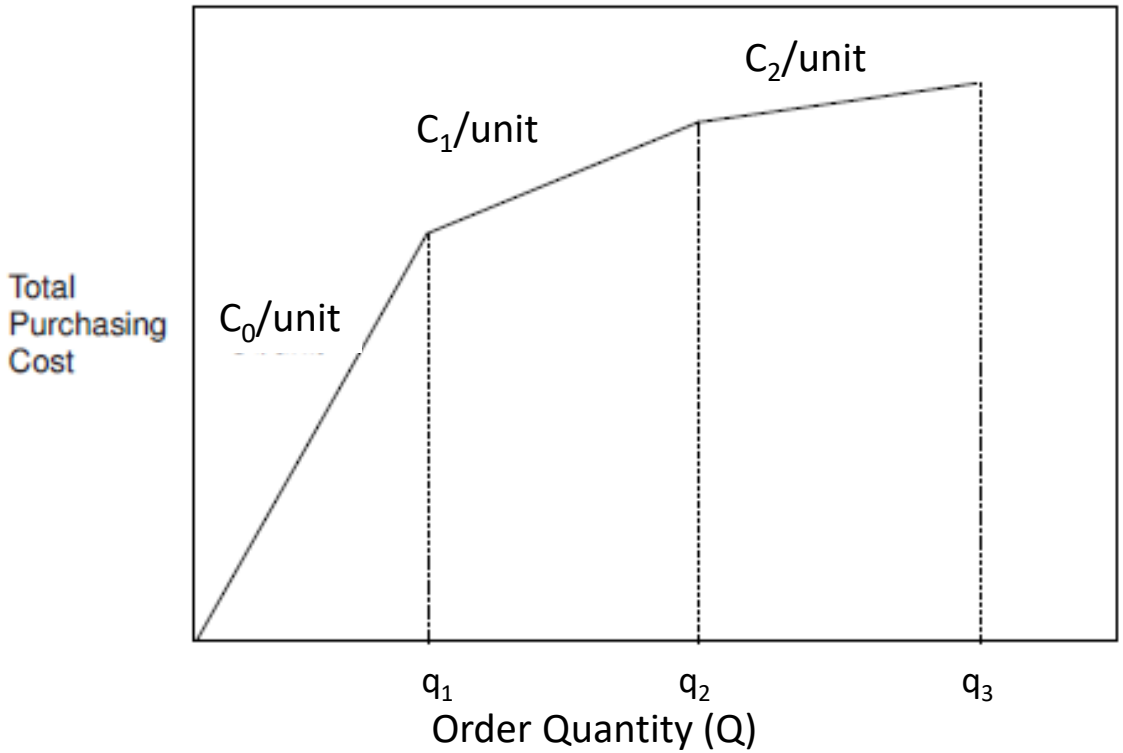
$$TVC_i = \frac{D}{q_i}S + \frac{q_i}{2}hC_i + DC_i$$

- Select the Q* with the lowest TVC

Marginal unit quantity discounts

- In this case, as the order quantity increases, the unit purchasing cost decreases only on the additional units purchased as opposed to on all the units purchased (that we discussed in the all unit discount scheme)
  - E.g., if the order Q is between q_i and q_{i+1}, only units over q_i are obtained at a cost C_i
- The objective for the company is to decide on a lot size that will minimize the sum of material, order, and holding costs
  - **TVC(Q) must now also include the purchasing cost**

Marginal unit quantity discounts: Purchasing cost



For $q_i \leq Q < q_{i+1}$:

$$C(Q) = C_0(q_1 - q_0) + C_1(q_2 - q_1) + \cdots + C_{i-1}(q_i - q_{i-1}) + C_i(Q - q_i)$$

Marginal unit quantity discounts: Per unit cost

- For $q_i \leq Q < q_{i+1}$, the purchasing cost is given by:

$$C(Q) = C_0(q_1 - q_0) + C_1(q_2 - q_1) + \cdots + C_{i-1}(q_i - q_{i-1}) + C_i(Q - q_i)$$

- Let R_i denote the sum of the terms in C(Q) that are independent of Q (notice that $R_0 = 0$ when $q_0 = 0$) :

$$R_i = C_0(q_1 - q_0) + C_1(q_2 - q_1) + \cdots + C_{i-1}(q_i - q_{i-1}) - C_i q_i$$

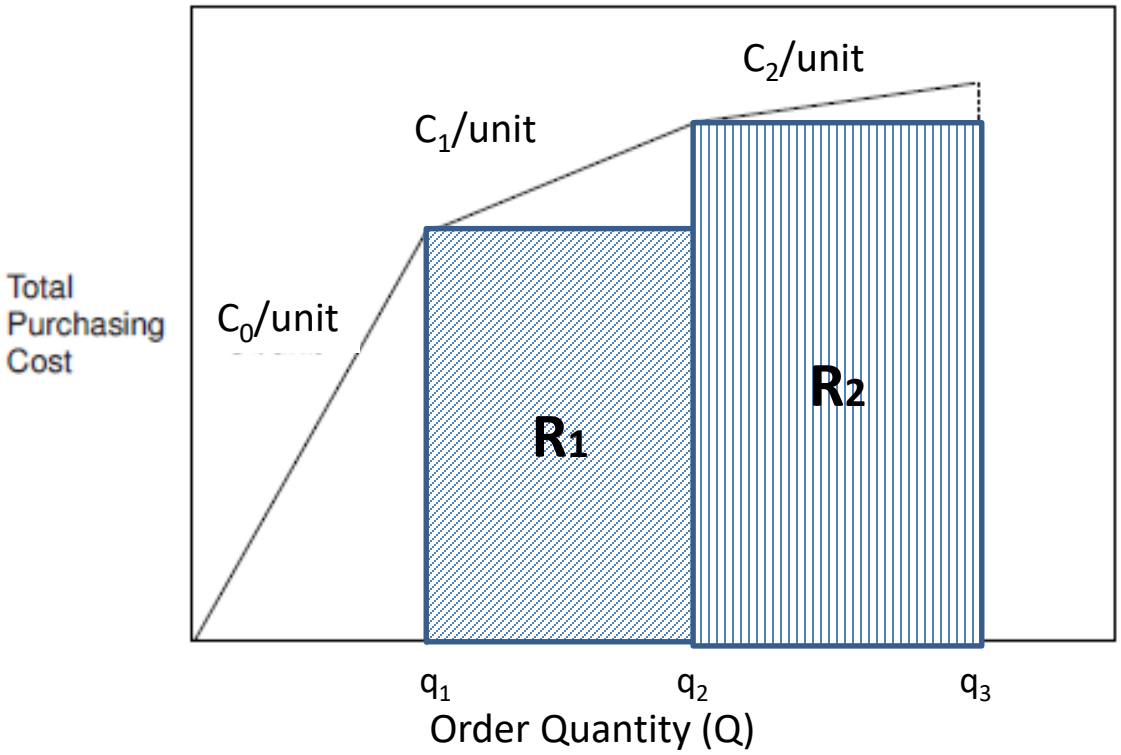
- Then, the total purchasing cost and the per unit cost are given by :

$$C(Q) = R_i + C_i Q$$

$$\frac{C(Q)}{Q} = \frac{R_i}{Q} + C_i$$

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Marginal unit quantity discounts: Purchasing cost



For $q_i \leq Q < q_{i+1}$: $C(Q) = R_i + C_i Q$

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Marginal unit quantity discounts: TVC functions

- For $q_i \leq Q < q_{i+1}$:

$$TVC_i(Q) = \frac{SD}{Q} + h\frac{C(Q)}{Q}\frac{Q}{2} + \frac{C(Q)}{Q}D$$

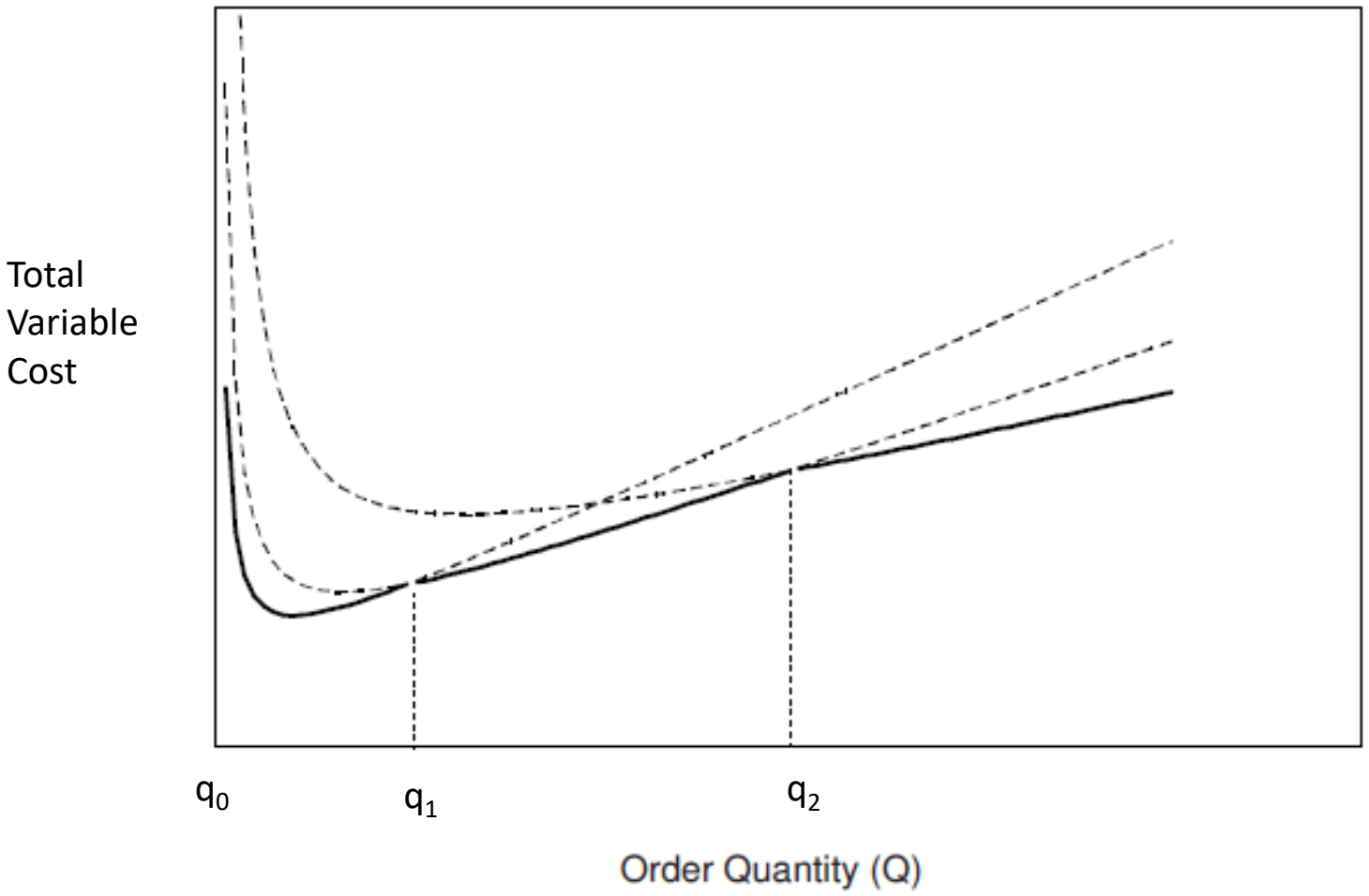
$$= \frac{(S + R_i)D}{Q} + h\frac{C_i Q}{2} + \left(C_i D + \frac{hR_i}{2}\right)$$

- Notice that there are *m* different TVC functions, one for each purchasing price (or quantity interval)

- By applying first order conditions on each TVC function, we can get the corresponding optimal quantities (which may or may not be *feasible*):

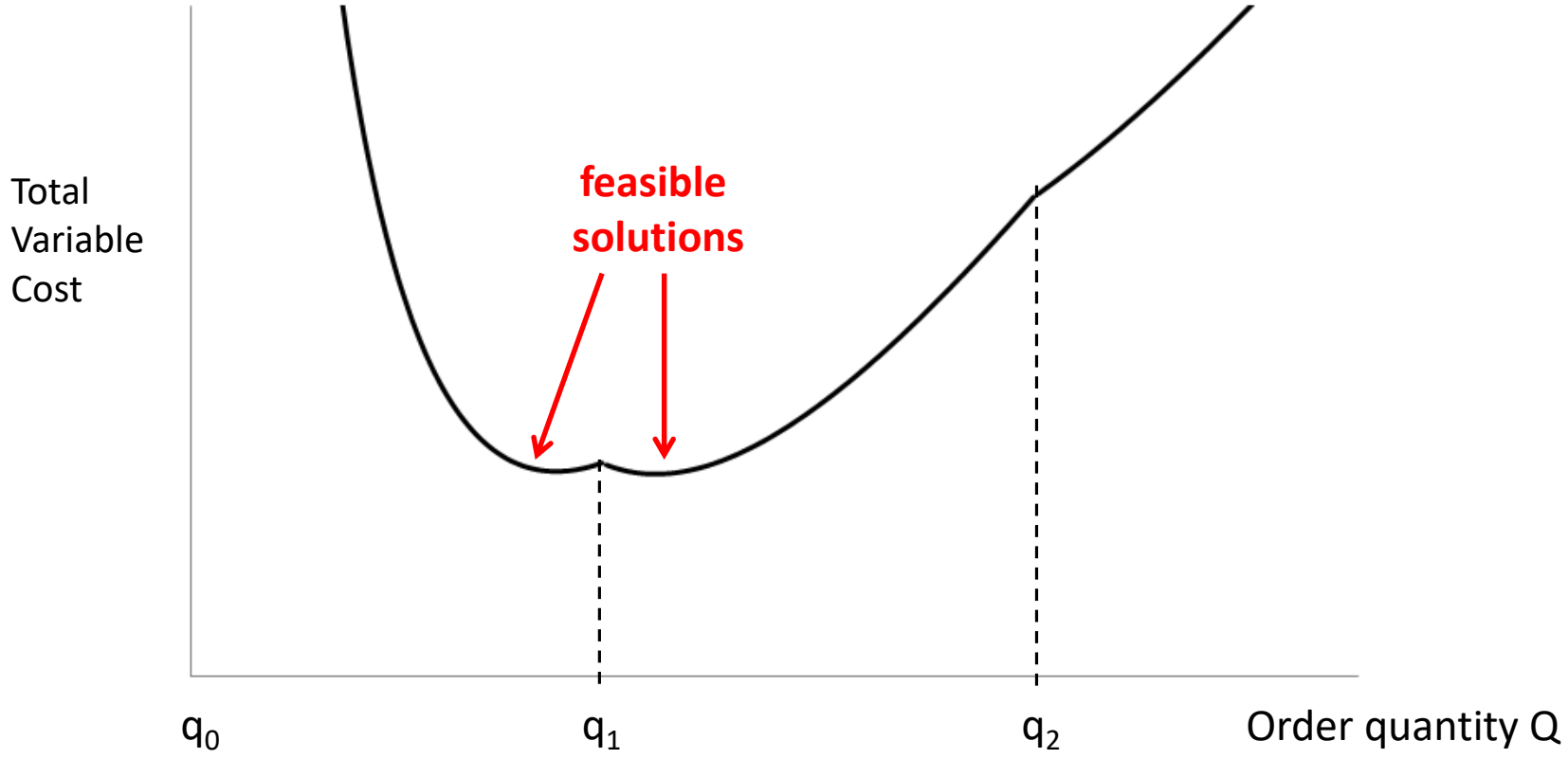
$$Q_i^* = \sqrt{\frac{2(R_i+S)D}{hC_i}}$$

Marginal unit discount: TVC vs. Q (1 feasible solution)



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Marginal unit discount: TVC vs. Q (2 feasible solutions)



Marginal unit quantity discounts: Solution algorithm

Step 1: Compute the order quantity (Q_i^*) that minimizes $TVC_i(Q)$ for each i (see previous slides). This step gives us a total of m possible order quantities.

Step 2: Check which of the potential Q_i^* values is feasible; i.e., $q_i \leq Q_i^* < q_{i+1}$. Disregard the non-feasible ones.

Step 3: Calculate the $TVC_i(Q_i^*)$ corresponding to each feasible Q_i^* value. The order quantity Q_i^* that produces the least cost is the EOQ.

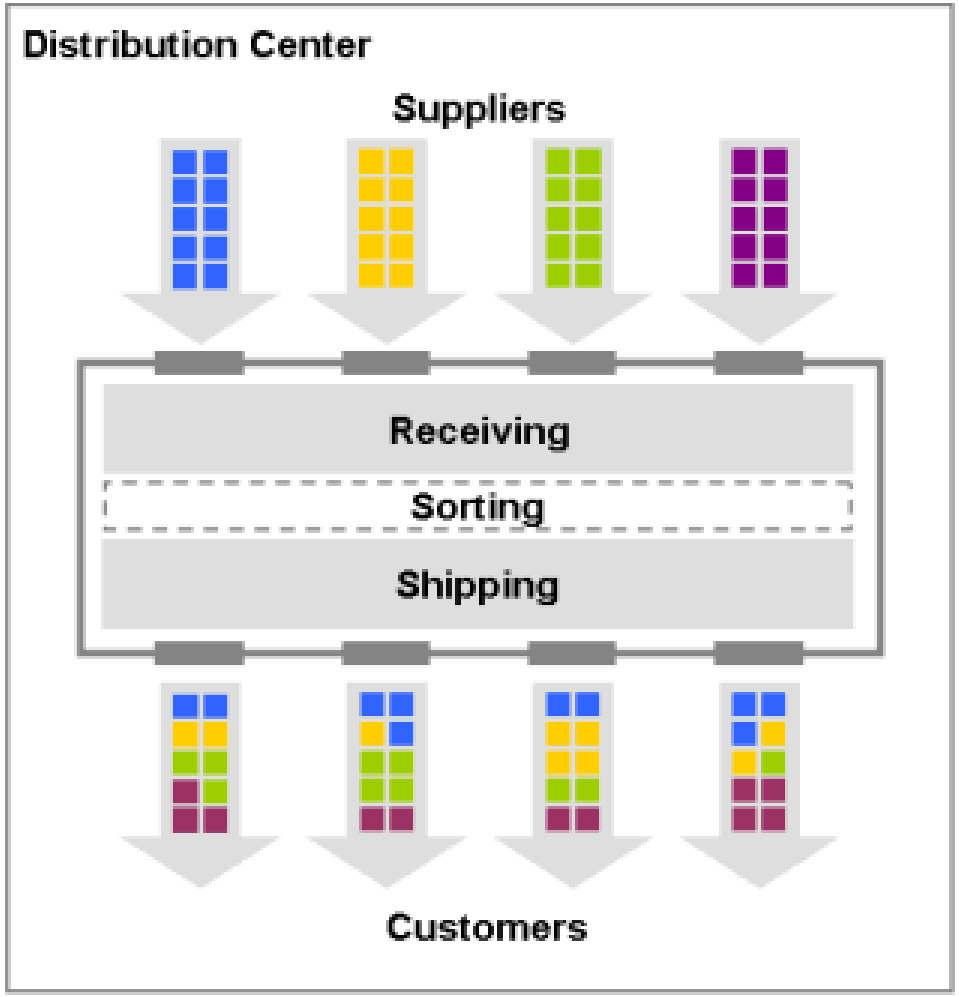
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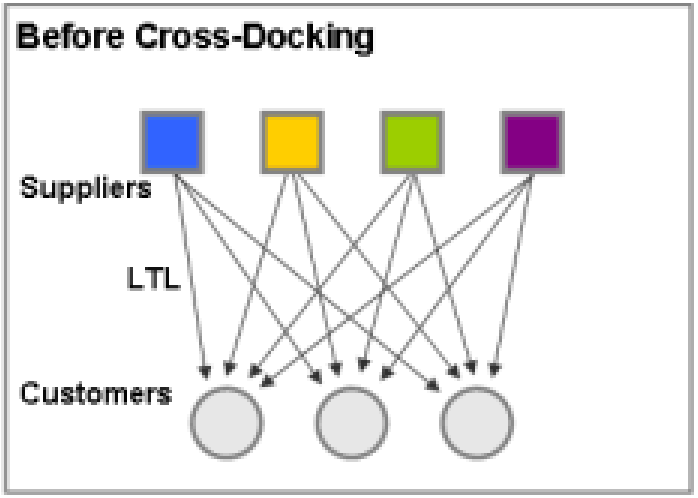
EOQ – AGGREGATION

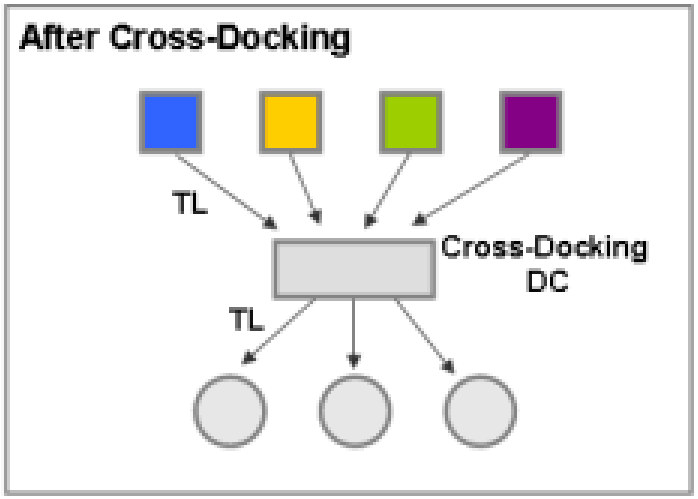
Aggregating multiple products in a single order

- Transportation is a significant contributor to the fixed cost per order.
- Can possibly combine shipments of different products from the same supplier.
- Can also have a single delivery coming from multiple suppliers or a single truck delivering to multiple retailers.
- Aggregating across products, retailers, or suppliers in a single order allows for a reduction in lot size for individual products because fixed ordering and transportation costs are now spread across multiple products, retailers, or suppliers.

Cross Docking







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Lot sizing with multiple products or customers

- Assume we have *m* products
  - Let S be the portion of the ordering cost that is independent of the variety of products ordered (e.g., transportation)
  - Let s_i be the product-specific ordering cost for product i, i = 1,…,m. This cost is dependent to the variety of products ordered (e.g., loading and receiving)
  - Denote by D_i , C_i and Q_i the annual demand, material cost and order quantity, respectively, for product i, i = 1,…,m
- **Delivery options**:
  - No Aggregation: Each product ordered separately
  - Complete Aggregation: All products delivered on each truck
  - Tailored Aggregation: Selected subsets of products on each truck

Complete aggregation: Order all products jointly

- Define: $S' = S + s_1 + s_2 + \cdots + s_m$
- Let n be the number of joint orders per year. It follows,

$$n = \frac{D_1}{Q_1} = \frac{D_2}{Q_2} = \cdots = \frac{D_m}{Q_m}$$

- Then, the total variable cost for complete aggregation is given by

$$TVC(n) = nS' + \frac{hC_1D_1}{2n} + \frac{hC_2D_2}{2n} + \cdots + \frac{hC_mD_m}{2n}$$

- By applying first order conditions on TVC(n), we get the optimal number of orders (n*):

$$n^* = \sqrt{\frac{\sum_i D_i h C_i}{2S'}} \quad \text{and} \quad Q_i^* = \frac{D_i}{n^*} \quad \text{for i = 1,…,m}$$

Tailored aggregation

- The weakness of complete aggregation is that low-demand products are aggregated with high-demand products
  - Results in high costs if the product-specific ordering cost for the low-demand products is large
- In this case, a tailored aggregation policy in which the low-demand products are ordered less frequently than the high-demand products may yield lower costs
  - For example, consider a policy in which high-demand products are included in every order, medium-demand products in every second order, and low-demand products in every fourth order (PO2 policy)
- Such policy is computationally more complex to derive (see Chopra & Meindl for an example)

Readings

- Chapter 10 (sections 10.1 to 10.4) from Chopra & Meindl
- Chapter 7 (sections 7.6 and 7.7) from Cachon & Terwiesch

Note: 10.2 in Chopra & Meindl includes a nice discussion on the different types of costs that are considered in the EOQ model

Thank you!